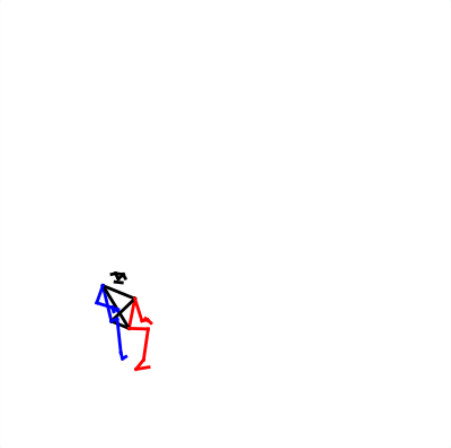
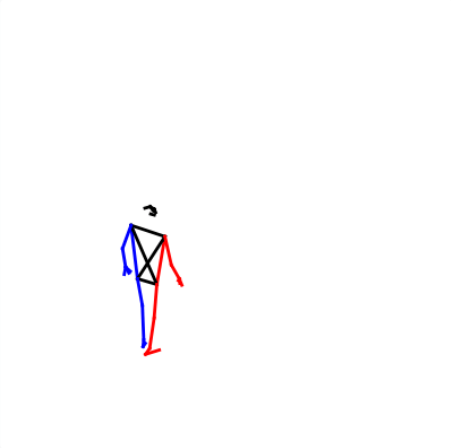
vailá - TUG Analysis Interactive Report

Subject: s26_m1_t1

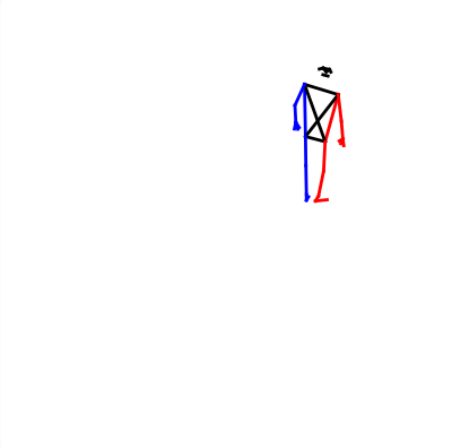
Phase Recordings



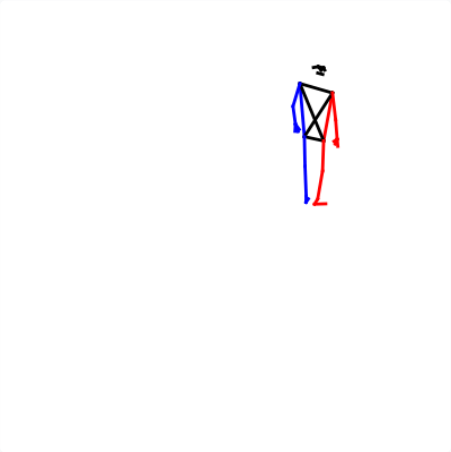
Stand



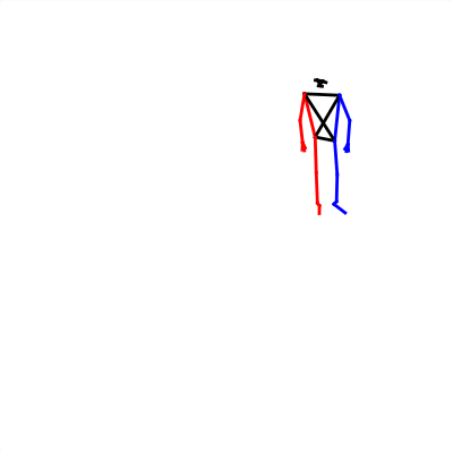
Gait Forward



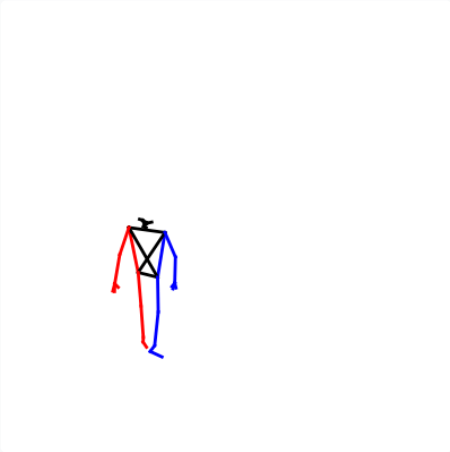
Stop 5s



Turn180

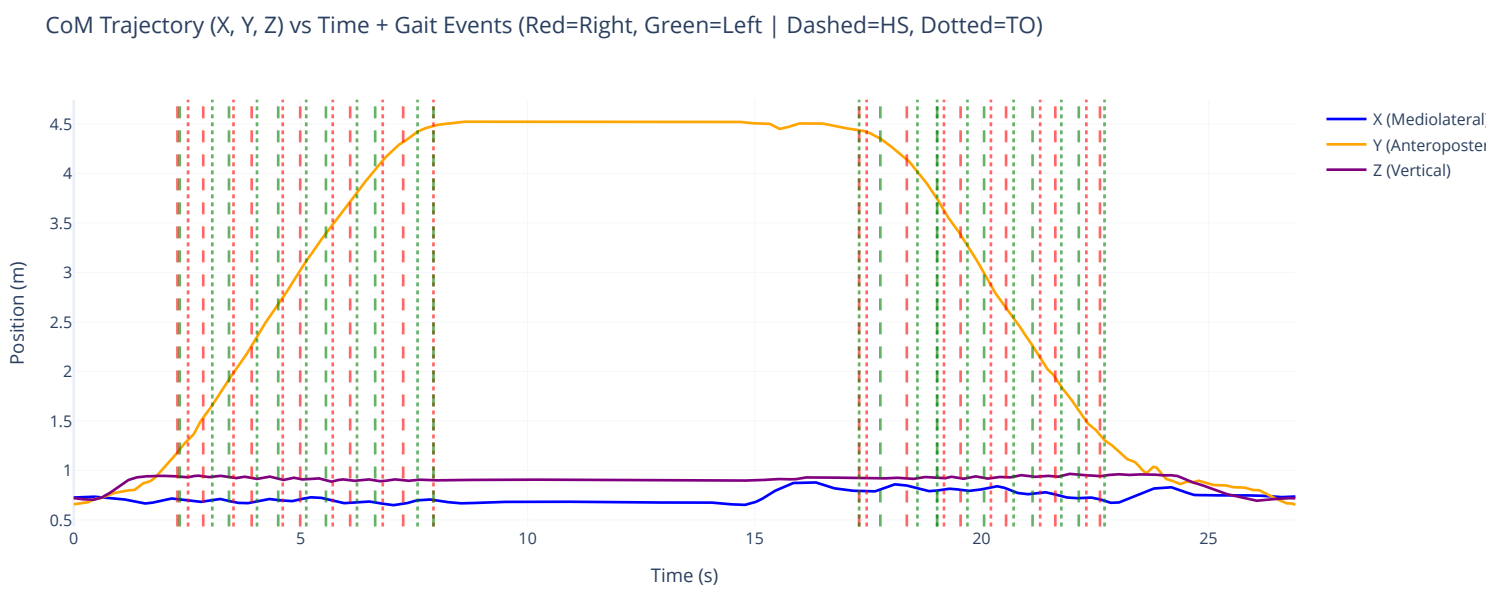


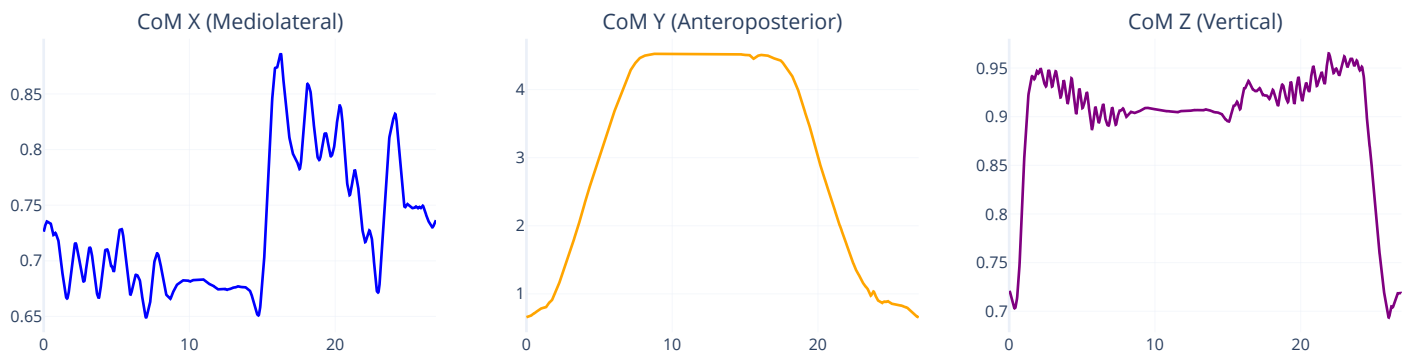
Gait Back

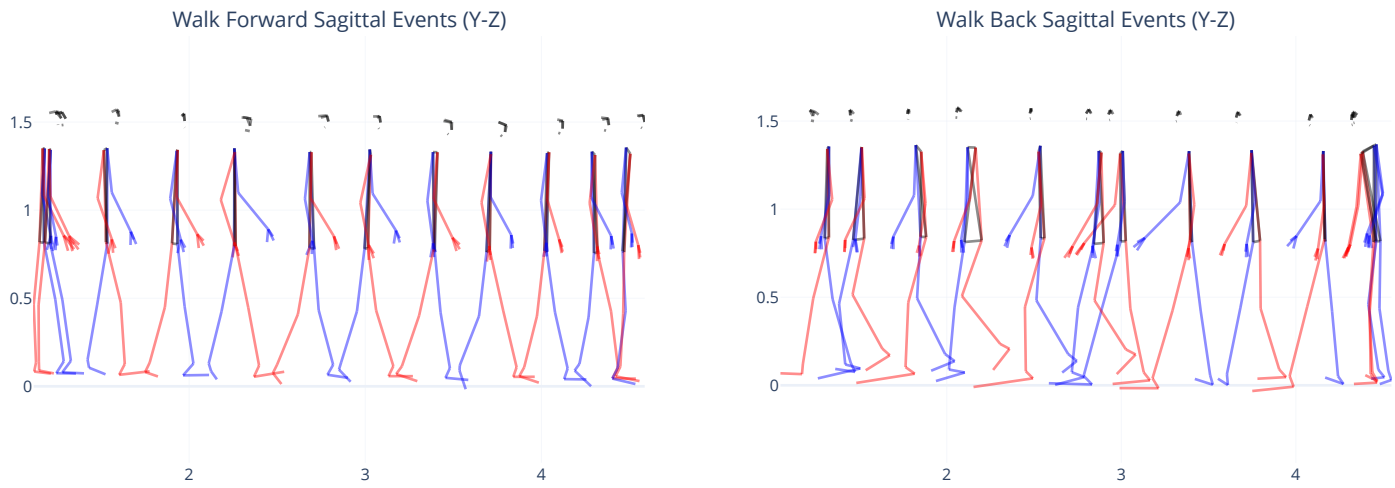


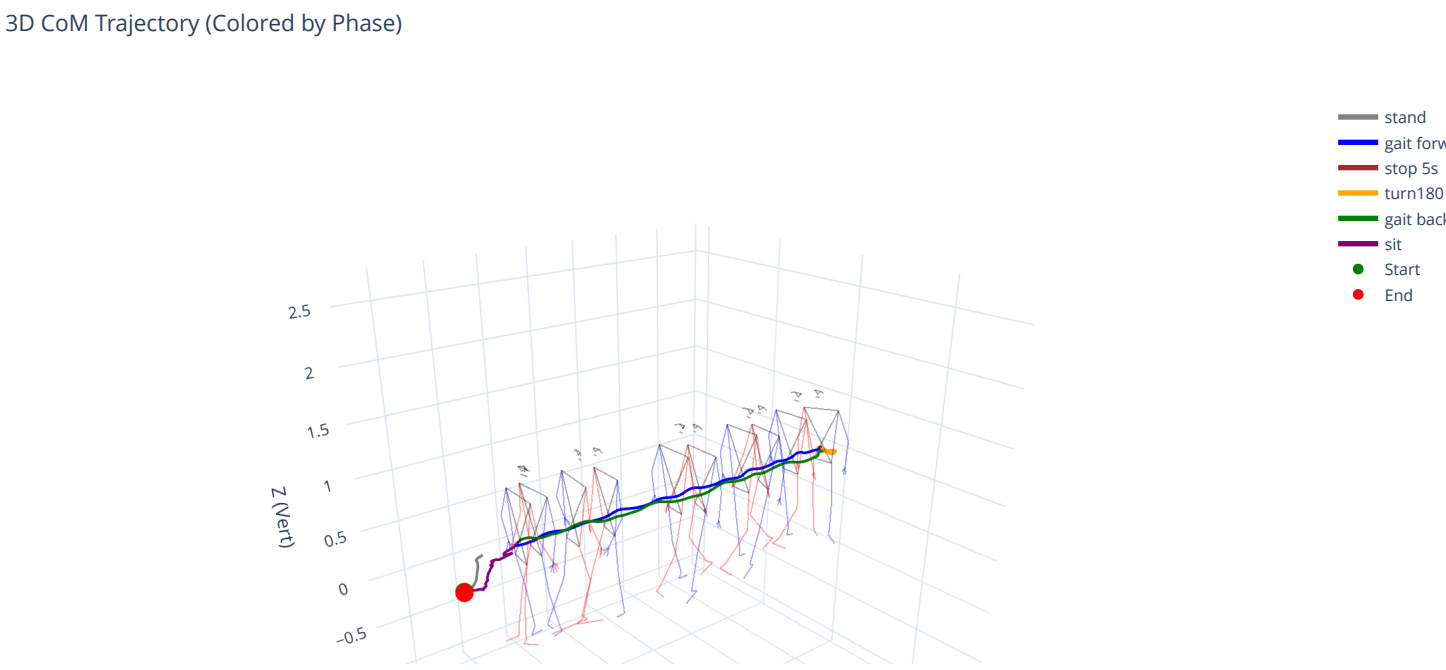
Sit

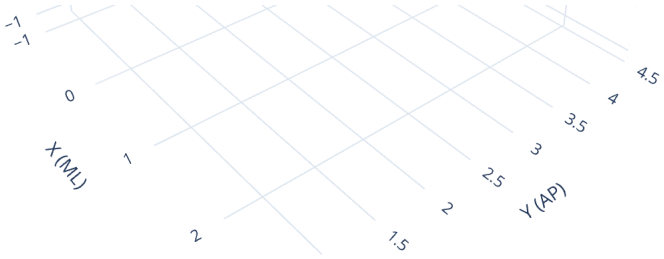
Visual Charts

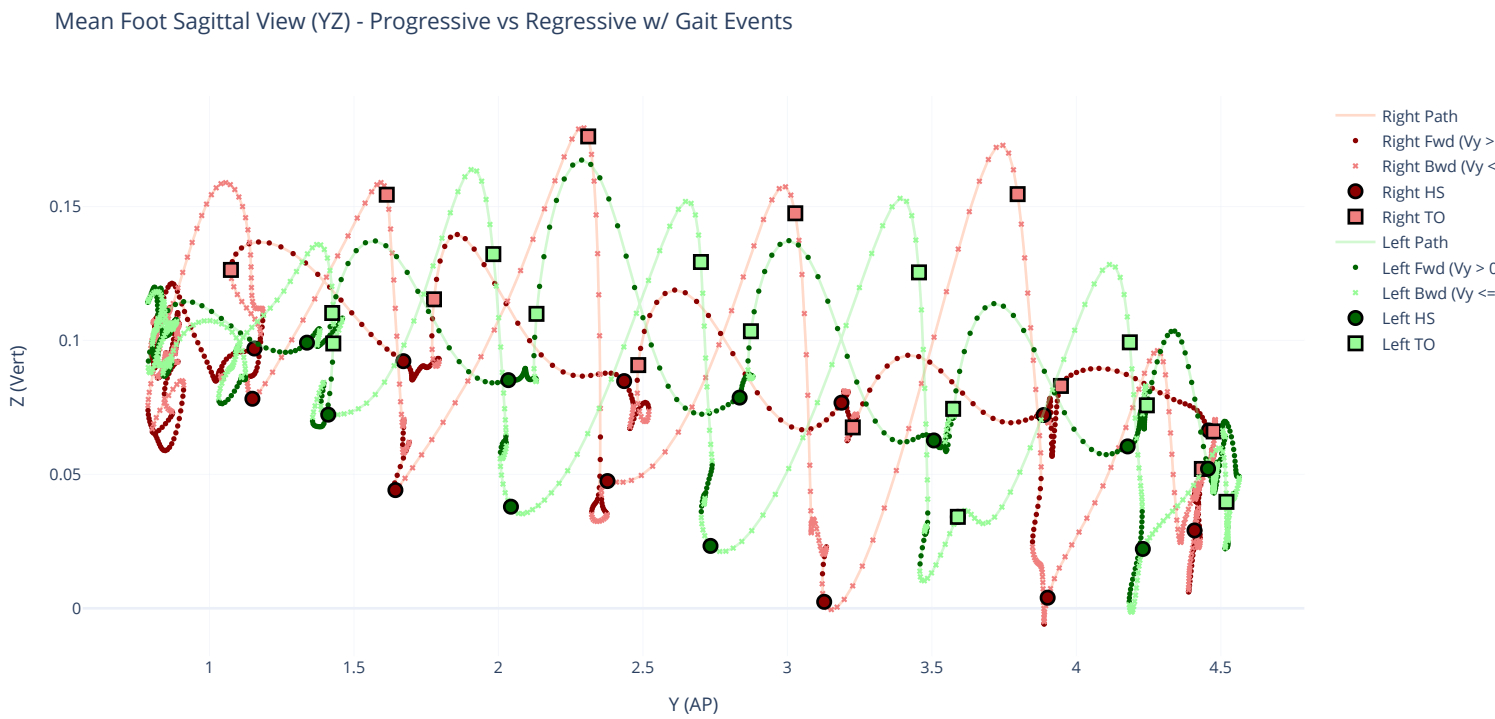


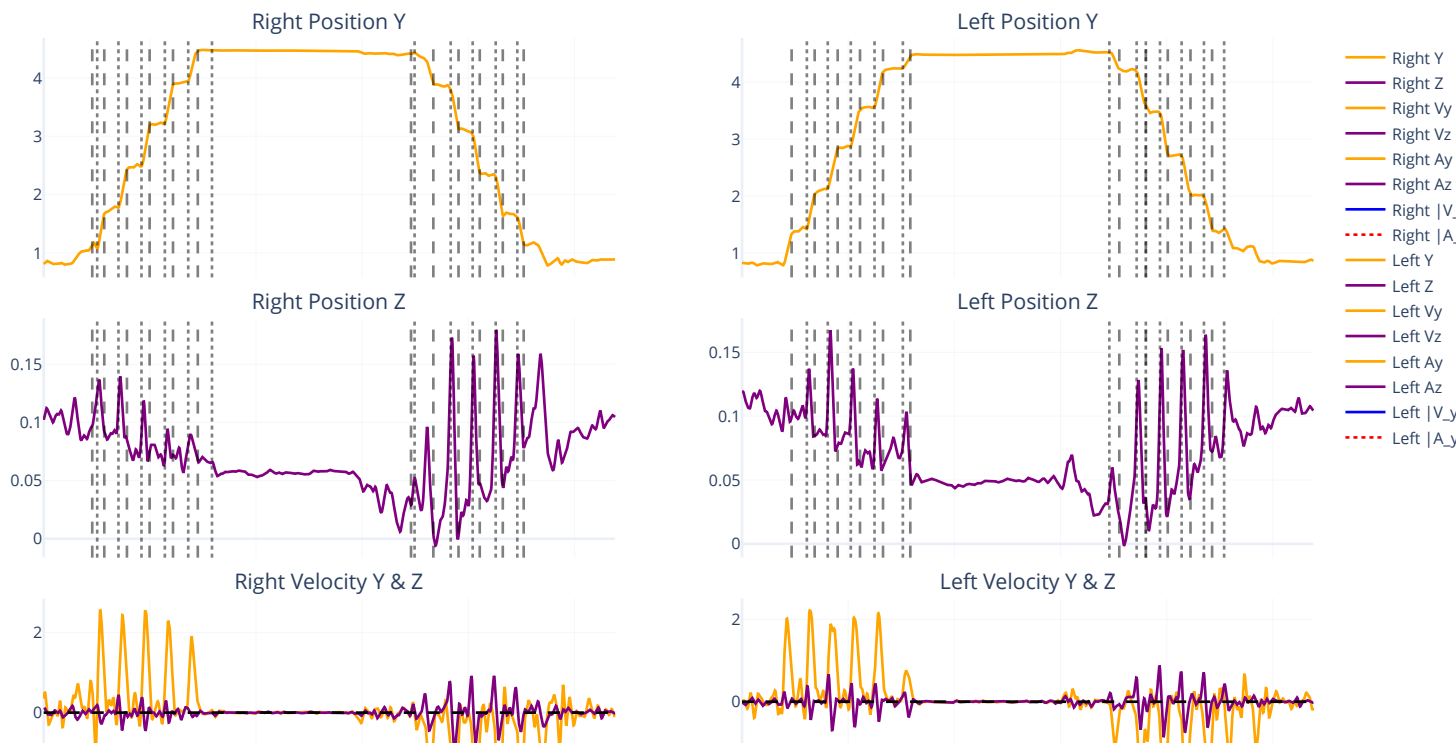

CoM X (Mediolateral)
CoM Y (Anteroposterior)
CoM Z (Vertical)

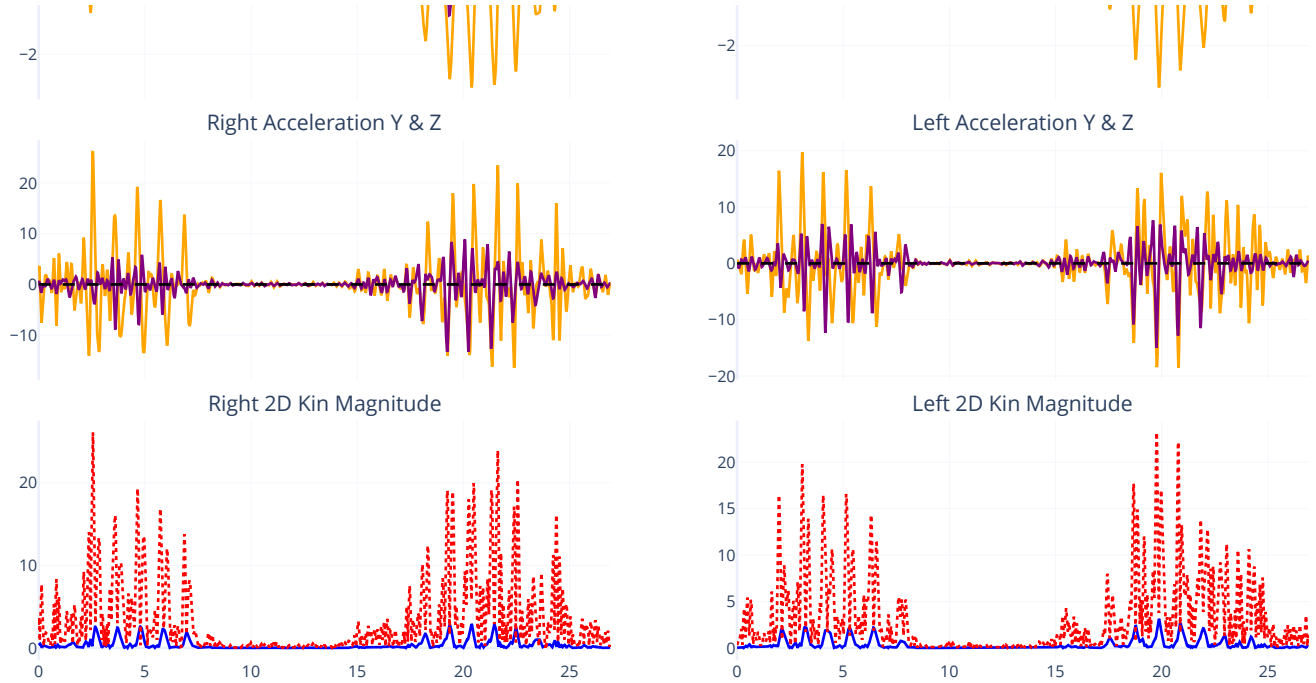

Walk Forward Sagittal Events (Y-Z)
Walk Back Sagittal Events (Y-Z)

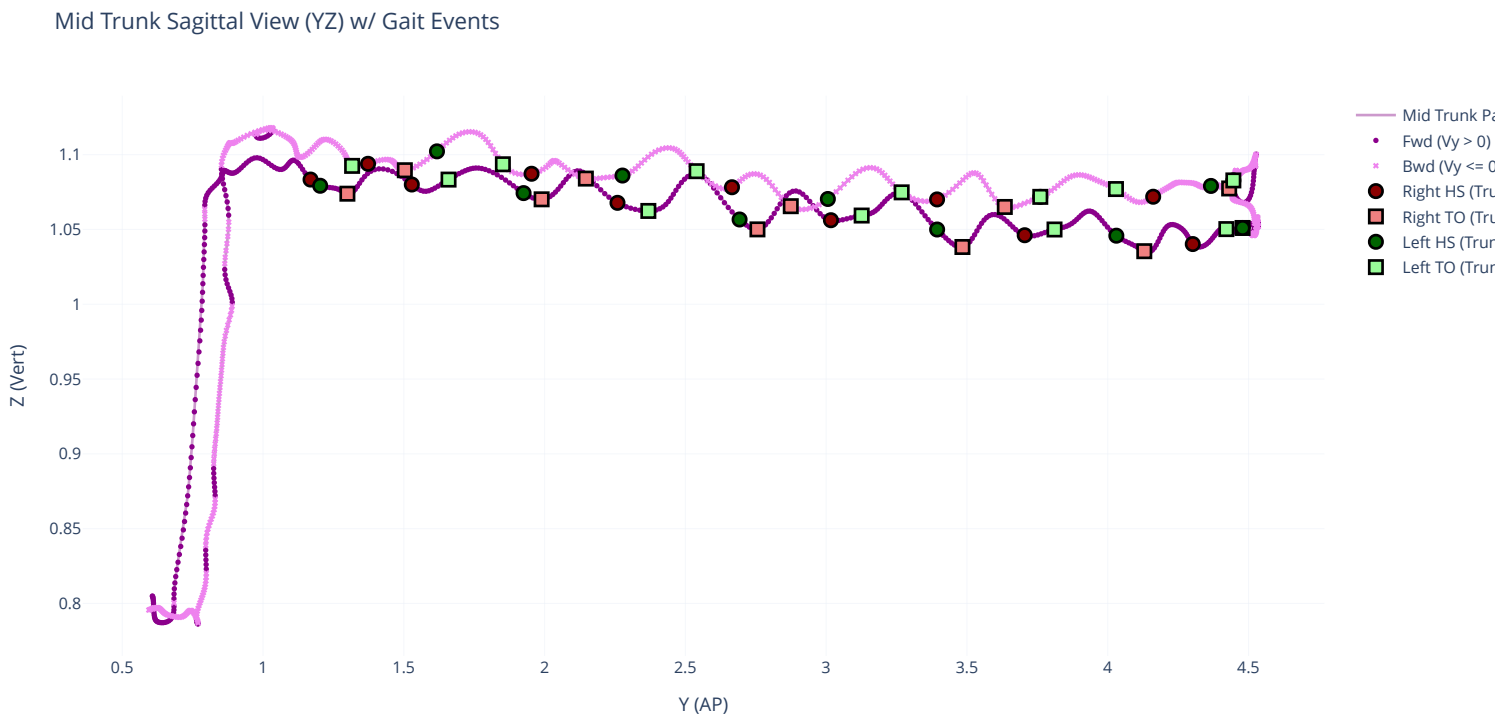

3D CoM Trajectory (Colored by Phase)
stand
gait forw
stop 5s
turn180
gait back
sit
Start
End
Z (Vert)

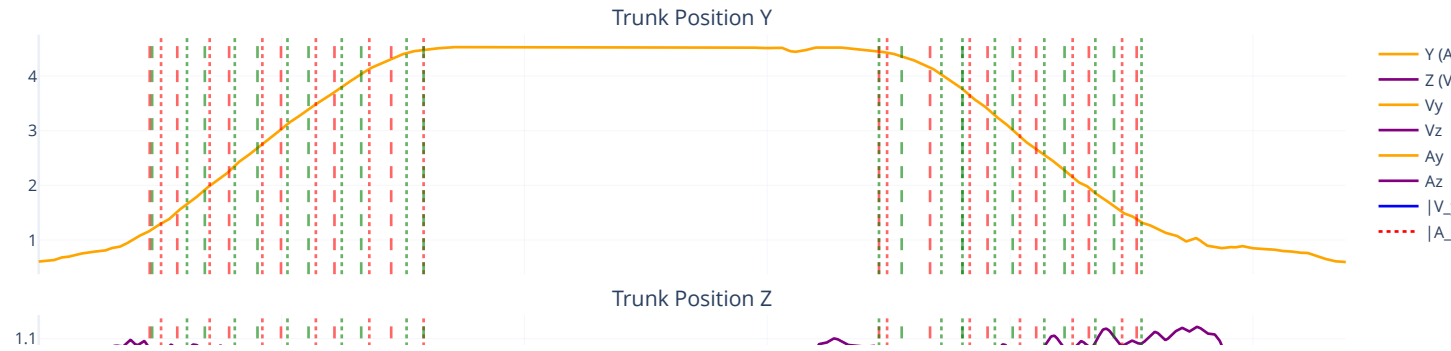

X (ML)
Y (AP)

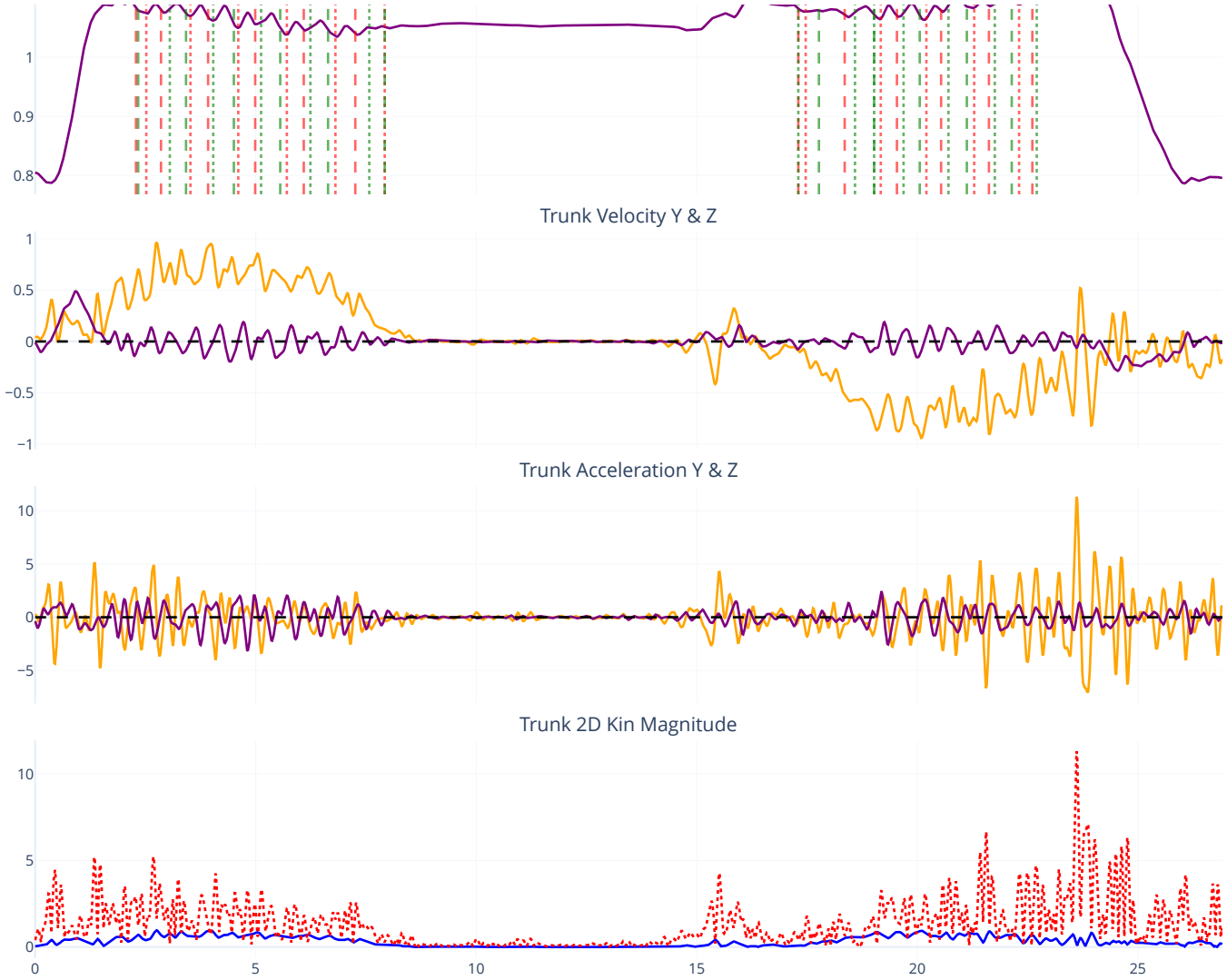

Mean Foot Sagittal View (YZ) - Progressive vs Regressive w/ Gait Events
Z (Vert)
Y (AP)
Right Path
Right Fwd (Vy >
Right Bwd (Vy <
Right HS
Right TO
Left Path
Left Fwd (Vy > 0
Left Bwd (Vy <=
Left HS
Left TO

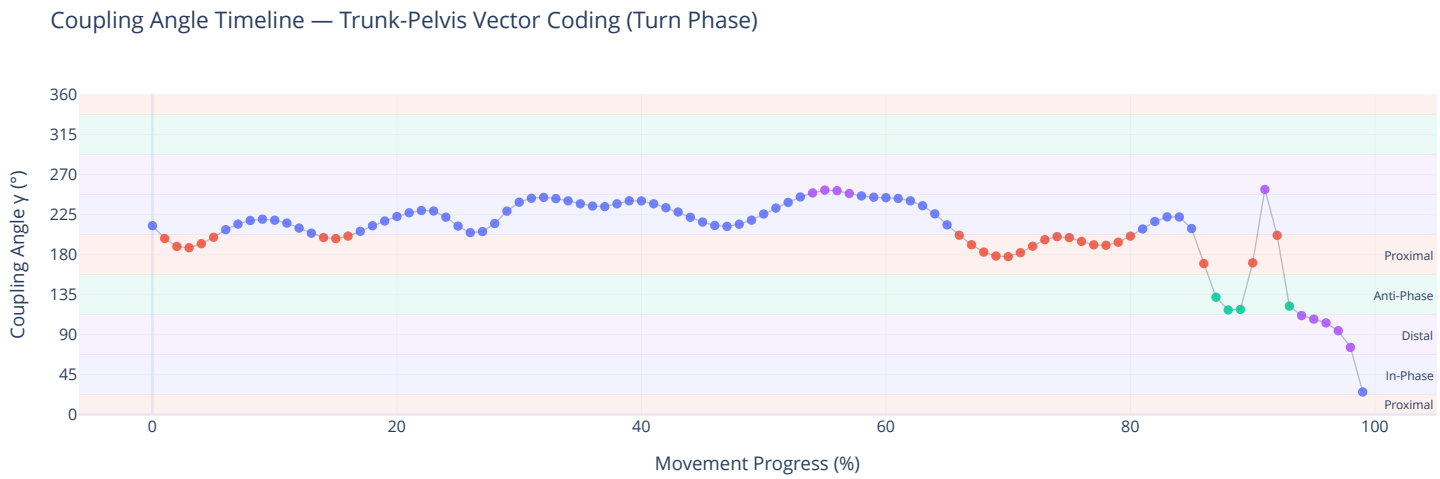

Mean Foot Kinematics Separated
Right Position Y
Left Position Y
Right Position Z
Left Position Z
Right Velocity Y & Z
Left Velocity Y & Z
Right Y
Right Z
Right Vy
Right Vz
Right Ay
Right Az
Right |V_
Right |A_
Left Y
Left Z
Left Vy
Left Vz
Left Ay
Left Az
Left |V_y
Left |A_y

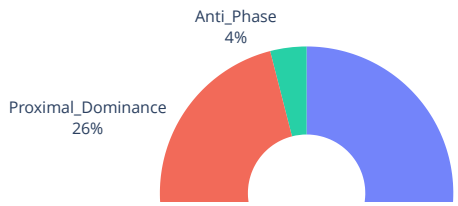

Right Acceleration Y & Z
Left Acceleration Y & Z
Right 2D Kin Magnitude
Left 2D Kin Magnitude

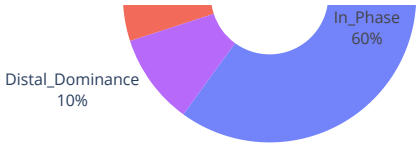

Mid Trunk Sagittal View (YZ) w/ Gait Events
Z (Vert)
Y (AP)
Fwd (Vy > 0)
Right HS
Right TO
Left HS
Left TO


Mid Trunk Kinematics Separated
Trunk Position Y
Trunk Position Z
Vy
Vz
Ay
Az

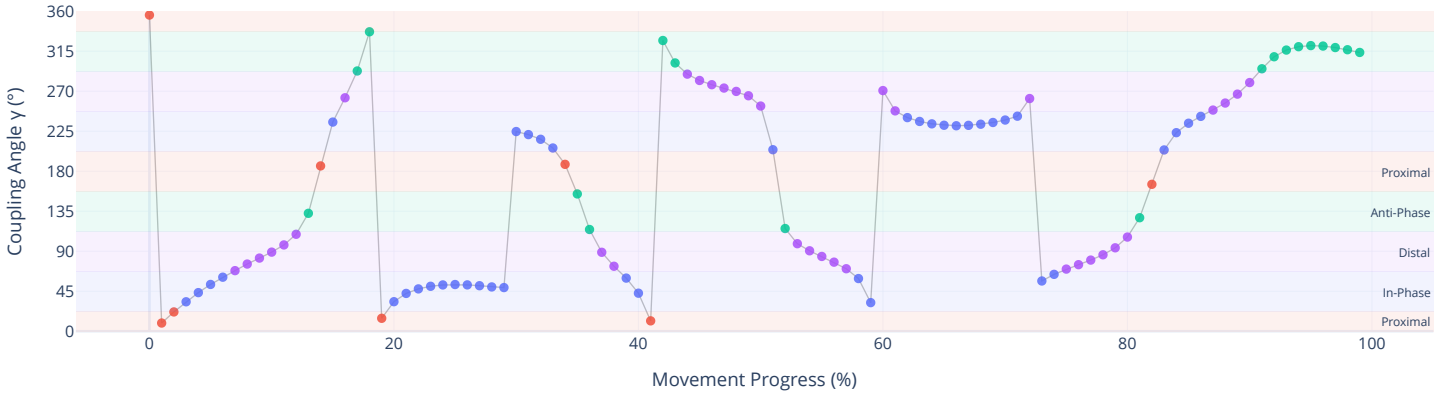

Trunk Velocity Y & Z
Trunk Acceleration Y & Z
Trunk 2D Kin Magnitude


Coupling Angle Timeline — Trunk-Pelvis Vector Coding (Turn Phase)
Coupling Angle γ (°)
Movement Progress (%)
Proximal
Anti-Phase
Distal
In-Phase
Proximal

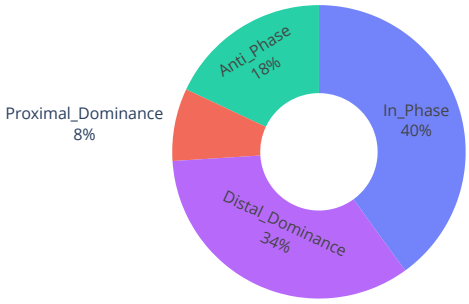

Coordination Pattern Distribution — Trunk-Pelvis Vector Coding (Turn Phase)
Anti_Phase
4%
Proximal_Dominance
26%



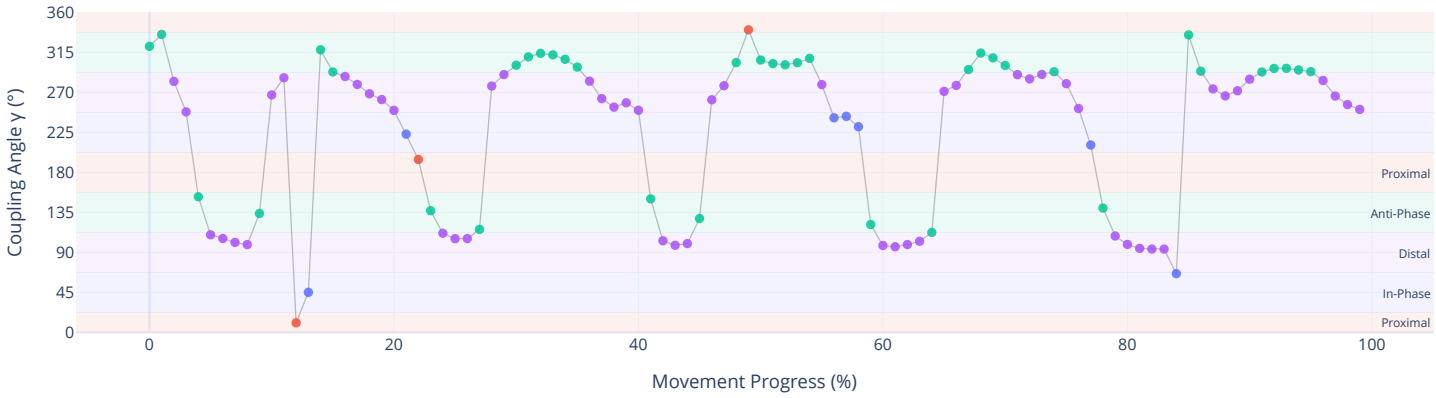
Coupling Angle Timeline — Trunk-Pelvis Vector Coding (Sit-to-Stand Phase)



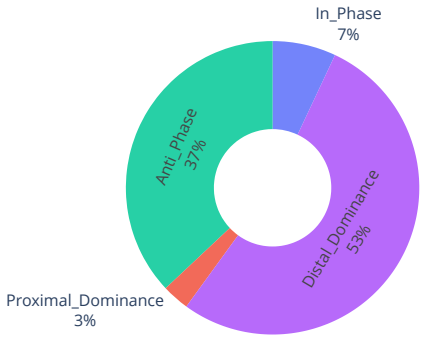
Coordination Pattern Distribution — Trunk-Pelvis Vector Coding (Sit-to-Stand Phase)



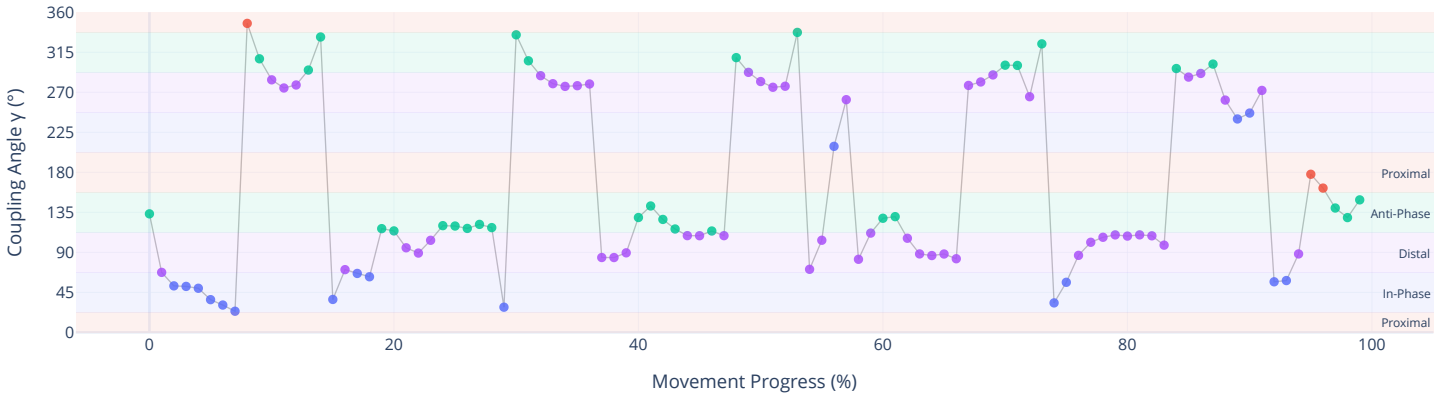
Coupling Angle Timeline — Arm-Leg Vector Coding (Walk Forward Analysis)



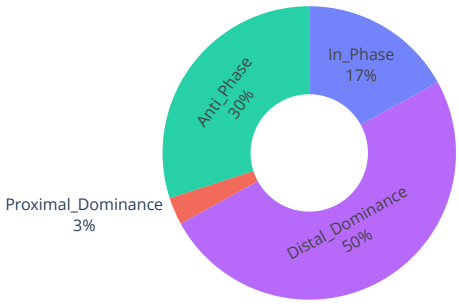
Coordination Pattern Distribution — Arm-Leg Vector Coding (Walk Forward Analysis)



Coupling Angle Timeline — Arm-Leg Vector Coding (Walk Back Analysis)



Coordination Pattern Distribution — Arm-Leg Vector Coding (Walk Back Analysis)



Metadata

Field	Value
SEX	M
WEIGHT	87.3
HEIGHT	1.63

Field	Value
AGE	63.0
FPS	59.94005994005994
STAGE_HY	3.0
PERFORMANCE	4
TURNSIDE	R

Spatiotemporal Parameters

Global

Variable	Value
Cadence steps per min	67.912
Velocity m s	0.347
XcoM Deviation Fwd m	0.027
XcoM Deviation Bwd m	0.041
Steps Walk Forward	12.000
Steps Walk Back	11.000
Fwd 1st strike inside zone	R
Fwd 1st complete step inside	L
Fwd Number of steps inside	11.000
Fwd Last strike inside zone	L
Fwd Sequence steps inside	['R', 'L', 'R', 'L', 'R', 'L', 'R', 'L', 'R', 'L', 'R', 'L']
Bck 1st strike inside zone	R
Bck 1st complete step inside	L
Bck Number of steps inside	10.000
Bck Last strike inside zone	R
Bck Sequence steps inside	['R', 'L', 'R', 'L', 'R', 'L', 'R', 'L', 'R', 'L', 'R']

Left vs Right

Variable	Left	Right
Stance Time s	0.691	0.694
Stance Time sd	0.104	0.029
Step Length m	0.321	0.368
Step Length sd	0.096	0.038
Step Width m	0.264	0.264
Step Width sd	0.077	0.063
Stride Length m	0.690	0.658
Stride Length sd	0.043	0.111
Swing Time s	0.409	0.365

Variable	Left	Right
Swing Time sd	0.040	0.043

Step-by-Step Timeseries (Gait Events / HS)

Step Index	Side	Time (s)	Y (m)	Phase	Stance Time (s)	Step Length (m)	Step Width (m)	Stride Length (m)	Swing Time (s)
1	Right	2.286	1.155	gait_forward	0.2336	0.5956	0.1528	0.4829	0.3337
2	Left	2.336	1.337	gait_forward	0.7174	0.1816	0.1575	0.686	0.367
3	Right	2.853	1.671	gait_forward	0.6673	0.3394	0.1905	0.7657	0.4004
4	Left	3.420	2.034	gait_forward	0.6173	0.3877	0.2439	0.8206	0.4671
5	Right	3.921	2.435	gait_forward	0.684	0.4186	0.2611	0.7642	0.3837
6	Left	4.505	2.835	gait_forward	0.6173	0.4243	0.3158	0.6699	0.4338
7	Right	4.988	3.187	gait_forward	0.7174	0.3612	0.2916	0.7095	0.3837
8	Left	5.556	3.507	gait_forward	0.684	0.3382	0.2903	0.6746	0.4004
9	Right	6.089	3.887	gait_forward	0.7174	0.3886	0.3068	0.5643	0.4505
10	Left	6.640	4.178	gait_forward	0.9343	0.3031	0.2651	0.2732	0.3504
11	Right	7.257	4.458	gait_forward	0.6673	0.2973	0.2315	0.0618	9.376
12	Left	7.925	4.456	gait_forward	9.376	0.1202	0.1066	0.3125	0.4671
13	Right	17.301	4.409	gait_back	0.1668	0.138	0.1511	0.4787	0.8842
14	Left	17.768	4.23	gait_back	0.8175	0.252	0.2024	0.6636	1.0344
15	Right	18.352	3.901	gait_back	0.8175	0.3179	0.31	0.7745	0.367
16	Left	19.019	3.59	gait_back	0.6673	0.3882	0.3423	0.7806	0.367
17	Right	19.536	3.127	gait_back	0.6673	0.4157	0.3243	0.7066	0.3337
18	Left	20.053	2.734	gait_back	0.6507	0.4221	0.3611	0.7134	0.4171
19	Right	20.537	2.377	gait_back	0.7508	0.3749	0.3233	0.7374	0.3337
20	Left	21.121	2.043	gait_back	0.634	0.4	0.3364	0.6386	0.3837
21	Right	21.622	1.643	gait_back	0.684	0.3907	0.3236	0.5956	0.3003
22	Left	22.139	1.411	gait_back	0.5672	0.3137	0.2791		
23	Right	22.606	1.148	gait_back		0.3742	0.3009		

TUG Phases (Seconds)

Phase	Range
stand	0.0s to 1.48s (Dur: 1.48s)
gait forward	2.17s to 7.97s (Dur: 5.81s)
stop 5s	7.97s to 14.51s (Dur: 6.54s)
turn180	14.51s to 17.3s (Dur: 2.79s)
gait back	17.3s to 23.06s (Dur: 5.76s)
sit	23.06s to 26.91s (Dur: 3.85s)

Phase	Range
Total TUG Time	26.926900000000003
Turn Direction	Right